

F1 Fantasy Team Optimization: A Contextual Linear Bandit Approach

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Abstract—Fantasy Formula 1 presents a sequential decision-making problem where participants select optimal racing teams under budget constraints to maximize points across a season. We formulate this as a contextual linear bandit problem, comparing two learning strategies: Standard Bandit, which updates only selected options, and Enhanced Bandit, which updates all options with full feedback. Using 70 races across 2023-2025 F1 seasons with 28 driver features and 18 constructor features, we demonstrate that Enhanced Bandit achieves 8,933 total points versus 8,585 for Standard Bandit and 8,485 for human baseline over 48 evaluation races. Enhanced Bandit shows consistent hyperparameter sensitivity with optimal alpha of 0.7 and lambda of 1.0, while Standard Bandit's performance is highly dependent on random seed with hyperparameter effects dominated by early random exploration. Feature importance analysis reveals recent performance metrics with weight of 4.27 and sprint race weekends with weight of 2.77 as dominant predictors. Our results demonstrate that contextual bandits successfully balance exploration-exploitation in fantasy sports, with Enhanced Bandit outperforming human decision-making by 5.3 percent through comprehensive learning, though deterministic selection may limit discovery of optimal team combinations.

Keywords—contextual bandits, reinforcement learning, sequential decision-making, fantasy sports, linear UCB

INTRODUCTION

Fantasy Formula 1 is a game where players build optimal racing teams by selecting 5 drivers and 2 constructors from a pool of 20 drivers and 10 constructors, while staying within a starting budget of \$100 million. The objective is to maximize points across a 24-race season, where points come from real race results such as finishing position, fastest lap, and overtakes. This problem becomes challenging since: (1) driver prices change each week based on performance, which makes budgeting difficult, (2) the player must lock in their team before the qualifying session, (3) the budget limit forces difficult trade-offs, and (4) driver performance changes over time due to team transfers and rules changes during or between seasons.

Fantasy sports are estimated to be a \$24+ billion global industry, with Formula 1 Fantasy alone attracting over 1 million players. This domain is ideal for testing decision-making algorithms because: (1) we have 46 features covering performance history, track characteristics, weather, and practice results, (2) points provide a clear measure of success after each race, (3) there is a natural trade-off between picking proven performers versus taking risks on uncertain drivers, and (4) we can compare algorithm performance against real human players.

This problem fits naturally into the contextual bandit framework: each race, we look at driver information, choose a team, and receive points. This work shows how contextual linear bandits can solve this problem while revealing which features matter most for predictions.

This work makes the following contributions:

- Application of contextual linear bandits to F1 Fantasy team selection with 46 engineered features
- Empirical comparison of Standard Bandit versus Enhanced Bandit learning strategies across 70 races
- Systematic hyperparameter optimization across 128 configurations showing strategy-specific optimal parameters
- Real-world validation demonstrating 5.3% improvement over human baseline (8,933 vs 8,485 points over 48 races)
- Feature importance analysis identifying recent performance and sprint weekends as dominant predictors
- Analysis showing Enhanced Bandit has consistent hyperparameter sensitivity while Standard Bandit performance is highly dependent on random seed

RELATED WORK

This work uses contextual bandits, specifically the Linear Upper Confidence Bound (LinUCB) algorithm [1, 2]. Contextual bandits solve sequential decision problems where you need to balance exploration (trying new things) with exploitation (sticking with what works), using available information to make better choices.

LinUCB works by maintaining confidence bounds on its predictions. For each option, it calculates a score combining the expected reward with an exploration bonus. The more uncertain the algorithm is about the option, the larger the bonus, encouraging it to try options it hasn't seen much. As more data comes in, the bonus shrinks and the algorithm converges to picking genuinely good options. This is different than simple approaches like epsilon-greedy that explore randomly.

Fantasy sports and sports prediction have been studied with various machine learning methods [3, 4]. The key difference here is treating team selection as a learning process rather than making fresh predictions each week. The algorithm gets better over time as it sees more race results. This captures the exploration-exploitation trade-off: selecting proven high-performers versus trying uncertain drivers who might surprise you.

We also compare two ways of learning. Standard Bandit only sees results for the options it actually selected, representing the classic online learning approach. Enhanced

Bandit sees everyone's results each race, using all available information to learn faster. Comparing these shows how much the extra information actually helps.

MODEL, ALGORITHM, AND EVALUATION

Problem Formalization

Contextual Linear Bandit Framework

At each race $t \in \{1, \dots, T\}$:

- *Context Observation*: Observe feature vectors $x_{i,t} \in \mathbb{R}^d$ for each option i where drivers have $d=28$ features and constructors have $d=18$ features.
- *Action Selection*: Select team $S_t = (D_t, C_t)$ where $D_t \subseteq \{1, \dots, 20\}$ with $|D_t| = 5$ (drivers) and $C_t \subseteq \{1, \dots, 10\}$ with $|C_t| = 2$ (constructors), subject to budget constraint.
- *Reward Observation*: Receive reward r_t = total fantasy points scored by team S_t .
- *Goal*: Maximize cumulative reward over all races.

Linear Reward Model

We assume expected points for option i at race t follows:

$$E[r_{i,t}|x_{i,t}] = \theta^T x_{i,t} \quad (1)$$

where $\theta \in \mathbb{R}^d$ is an unknown weight vector learned from observations.

LinearUCB Algorithm

For each option i , maintain:

- $A_i \in \mathbb{R}^{d \times d}$: Covariance matrix, initialized as

$$A_i = \lambda I \quad (2)$$

- $b_i \in \mathbb{R}^d$: Reward accumulator, initialized as

$$b_i = 0 \quad (3)$$

At each race t :

- *Step 1 - Compute Weight Estimates*:

$$\theta_i = A_i^{-1} b_i \quad (4)$$

- *Step 2 - Calculate UCB Scores*:

$$UCB_{i,t} = \theta_i^T x_{i,t} + \alpha \sqrt{(x_{i,t}^T A_i^{-1} x_{i,t})} \quad (5)$$

where α controls exploration and the second term captures uncertainty.

- *Step 3 - Team Selection*: Select team S_t maximizing sum of UCB scores subject to budget and size constraints.
- *Step 4 - Update*: After observing rewards $r_{i,t}$, update:

$$A_i = A_i + x_{i,t} x_{i,t}^T \quad (6)$$

$$b_i = b_i + r_{i,t} x_{i,t} \quad (7)$$

Two Learning Strategies

- *Standard Bandit*: Updates A_i and b_i only for selected team S_t (7 observations per race). True online learning with limited feedback
- *Enhanced Bandit*: Updates A_i and b_i for all drivers and constructors (30 observations per race). Faster learning from comprehensive data

Hyperparameters

- α (*exploration*): Controls confidence bound width. Higher α = more exploration.
- λ (*regularization*): Ridge penalty on θ . Higher λ = less overfitting.

Feature Engineering

Driver Features (28 total)

- *Performance History (9)*:
 - Recent form (weighted average last 3 races: 0.6, 0.8, 1.0)
 - Overall average (exponential decay rate=0.95 over 10 races)
 - Track specific performance
 - Performance trend
 - Consistency (standard deviation)
 - Max/min points
 - Negative points rate
 - Upside potential
- *Weekend Signals (1)*:
 - Sprint weekend indicator
- *Driver Attributes (4)*:
 - Experience (GP entries), age, championships, podiums
- *Cost dynamics (3)*:
 - Current cost, average cost change over last 3 races, value (points per cost)
- *Environmental (7)*:
 - Track characteristics (length, turns, DRS zones, type)
 - Weather (max/min temperature, rain probability)
- *Driver of the Day (3)*:
 - DOTD win rate (last 10 races), recent DOTD streak (last 3 races), total DOTD wins
- *Bias (1)*:
 - Bias term

All features normalized to appropriate ranges for numerical stability.

Constructor Features (18 total)

- *Performance History (9)*:
 - Recent form (weighted average last 3 races: 0.6, 0.8, 1.0)
 - Overall average (exponential decay rate=0.95 over 10 races)
 - Track specific performance
 - Performance trend
 - Consistency (standard deviation)
 - Max/min points
 - Hot Streak (recent vs historical average)
 - Upside potential

- *Weekend Signals (1)*:
 - Sprint weekend indicator
- *Cost dynamics (3)*:
 - Current cost, average cost change over last 3 races, value (points per cost)
- *Environmental (4)*:
 - Track characteristics (length, turns, DRS zones, type)
- *Bias (1)*:
 - Bias term

All features normalized to appropriate ranges for numerical stability.

Design Rationale: The feature set balance (1) recency weighting (recent races weighted more heavily), (2) track conditioning (teams perform differently at each circuit), and (3) weekend-specific signals (sprint weekends provide different racing formats).

Team Selection

The optimal team selection problem is computationally hard (bounded knapsack variant). Rather than using a greedy heuristic, we use an exhaustive search with pruning. The algorithm tries all combinations of 5 drivers and 2 constructors, checking budget constraints and tracking the best team found. To reduce computation, we prune branches where the upper bound on possible value cannot exceed the current best. The highest-UCB driver in the selected team receives a $2\times$ multiplier (F1 Fantasy rule). While this exhaustive approach has worst-case complexity of $O(C(20,5) \times C(10,2)) = O(15,504 \times 45)$ combinations, pruning significantly reduces the actual search space in practice.

Experimental Setup

Dataset: F1 Seasons 2023-2025: $22 + 24 + 24 = 70$ total races. Evaluation on 48 races (2024-2025). Real data including costs, points, practice results, sprint weekends, DOTD awards.

Evaluation: Online learning across 70 races. **Metrics:** (1) Total points (cumulative fantasy score), (2) Average points/race, (3) Performance vs human baseline.

Baselines: Human (author’s actual selections), Oracle (perfect hindsight), Greedy (top points/cost ratio), Random (random valid team).

Hyperparameter Tuning: Grid search: $\alpha \in \{0.1, 0.2, 0.3, 0.4, 0.5, 0.6, 0.7, 0.8\}$, $\lambda \in \{0.5, 1.0, 1.5, 2.0, 2.5, 3.0, 3.5, 4.0\}$, Strategy $\in \{\text{Standard, Enhanced}\}$. Total: 128 configurations.

TABLE I. ALGORITHM PERFORMANCE COMPARISON

Method	Total	Avg/Race	vs Human
Enhanced Bandit	8933	186.1	+448 (+5.3%)
Standard Bandit	8585	178.9	+100 (+1.2%)
Human	8485	176.8	-
Oracle	14431	300.6	+5946 (+70%)
Greedy	6272	130.7	-2213 (-26%)
Random	4529	94.4	-3956 (-47%)

RESULTS

Main Results

Table I shows performance over 48 evaluation races (2024-2025).

Insights:

Enhanced Bandit beats human baseline by 448 points (5.3%)

Standard Bandit slightly beats human by 100 points (1.2%)

Enhanced outperforms Standard by 348 points (4.1%)

Both significantly exceed simple baselines, validating the contextual bandit framework

Season Breakdown

Table II shows that all methods improve from 2024 to 2025. Enhanced Bandit maintains consistent advantage across different grid conditions.

Hyperparameter Analysis

Optimal Parameters:

Enhanced Bandit: $\alpha = 0.7, \lambda = 1.0$

Standard Bandit: $\alpha = 0.7, \lambda = 3.5$

Insights: Enhanced Bandit shows consistent hyperparameter behavior across runs. Changes to alpha and lambda produce predictable effects on performance.

Standard Bandit behaves differently. Optimal hyperparameters vary substantially with different random seeds. The reported parameters ($\alpha=0.7, \lambda=3.5$) are from one run and may not generalize to other seeds.

This instability comes from Standard Bandit’s limited feedback (7 observations per race). Early random exploration decisions spread through the season, and this effect dominates the influence of hyperparameter tuning.

Feature Importance

We calculate feature importance by combining the learned weight vectors across all drivers.

Low-Weight Features ($|\theta| < 0.5$): Upside potential (0.47), performance trend (0.44), circuit length (0.40), experience (0.39), recent DOTD win (0.29), championships (0.29), max temp (0.16), min temp (0.13), min points (0.10), track-specific performance (0.09), consistency (0.05), max points (0.01), number of DOTD wins (0.01).

Insights:

Performance history dominates (Recent Form 4.27 + Overall Average 2.44 = 24% of total weight)

Weekend Signals provide useful information (Sprint Race weekend is second most important at 2.77)

Surprising findings: (1) DOTD win rate matters (2.14) but total DOTD wins doesn’t (0.01), and (2) Track-specific performance surprisingly low (0.09) despite intuition.

TABLE II. PERFORMANCE BY SEASON

Method	2024	2025	Change
Enhanced Bandit	4368	4565	+4.5%
Standard Bandit	4210	4375	+3.9%
Human	4148	4337	+4.6%

Variance Analysis

- *Best run:* 8,969 points (beats Enhanced).
- *Worst run:* 7,948 points
- *Range:* 1,021 points (11.9%)

Enhanced Bandit is deterministic.

This variance comes from the random exploration in early races affecting the entire season. Different early driver picks lead to different learning paths and final results. This randomness is so strong that it eliminates the effect of hyperparameters. The best alpha and lambda values for Standard Bandit vary wildly across different runs, making the algorithm hard to tune consistently.

DISCUSSION

Why Enhanced Bandit Outperforms Standard Bandit

Information Advantage: Enhanced sees 30 observations per race. Standard only sees 7. This $4.3\times$ more data helps Enhanced learn faster and converge to better predictions.

Variance Reduction: Enhanced is more consistent because it learns from all the data. Standard is sensitive to which specific drivers it happens to pick. As stated previously, this randomness is so strong that it completely dominates hyperparameter choice. Enhanced has a clear relationship between alpha, lambda, and performance that you can tune. Standard's best hyperparameters change unpredictably between runs, making it fundamentally difficult to optimize.

However, Enhanced Bandit's determinism creates a problem. It might get stuck always picking decent teams without ever finding better combinations. Standard's randomness occasionally stumbles onto better solutions that Enhanced never explores.

Feature Engineering Impact

What Worked:

- Recency weighting (exponential decay rate=0.95, weighted averages 0.6/0.8/1.0) captures form changes

TABLE III. TOP 10 DRIVER FEATURES BY ABSOLUTE WEIGHT

Ran k	Feature	$ \theta $
1	Recent Form (3 races)	4.27
2	Sprint Race Weekend	2.77
3	Overall Average	2.44
4	DOTD Win Rate	2.14
5	DRS Zones	1.92
6	Cost Change	1.51
7	DNF Rate	1.49
8	Number of Turns	1.40
9	Podiums	1.21
10	Track Type	1.20

- Sprint weekend indicator provides useful signal
- Track-specific conditioning adapts to circuit characteristics

What Didn't Work:

- Weather features too noisy (temperature, rain)
- Track-specific performance had minimal impact despite intuition that past performance at the same track would be predictive

Limitations

Deterministic Enhanced Bandit: No randomization may prevent discovery of better combinations.

High dimensionality: 46 features may overfit. Many low-weight features contribute minimally.

Single human baseline: Evaluated against one human player (author). Multi-user validation needed.

CONCLUSION AND FUTURE DIRECTIONS

We successfully applied contextual linear bandits to F1 Fantasy team optimization. Enhanced Bandit achieved 8,933 points over 48 races, beating human baseline by 448 points (5.3%) and Standard Bandit by 348 points (4.1%). Through systematic hyperparameter tuning, we found Enhanced Bandit has consistent, tunable hyperparameter sensitivity (optimal $\alpha=0.7$, $\lambda=1.0$), while Standard Bandit's performance is dominated by random seed effects that overwhelm hyperparameter tuning.

Feature importance analysis revealed recent performance (weight=4.27) and sprint race weekends (weight=2.77) as dominant predictors, with 17/46 features contributing minimally ($|\theta| < 0.5$). Notably, track-specific historical performance had very low weight (0.09), suggesting that recent form matters more than past performance at the same circuit.

Key Takeaways:

- Contextual bandits effectively balance exploration-exploitation in fantasy sports
- Comprehensive data helps when available, but requires careful regularization
- Domain knowledge essential for effective feature engineering. Understanding F1 enabled identification of predictive signals like sprint weekends and practice sessions

Algorithmic Improvements:

- *Add epsilon-greedy to Enhanced Bandit:* to escape potential local optima
- *Strategic budget planning:* predict cost changes to balance current points with future budget building
- *Alternative approaches:* Dueling Bandits (direct driver pair comparisons), Combinatorial Bandits (optimize full team as single action)

Feature Engineering:

- *Dimensionality reduction:* remove 17 features with $|\theta|<0.5$, reducing from 46 to 29 features
- Driver transfer indicators for new team adaptation

Real-World Deployment:

- Deploy for 2026 season to validate live performance against evolving human strategies.

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